

From the Received RF Signal to I/Q Samples

and how the carrier frequency offset (CFO) is measured in a BLE 5.1 CTE packet

BLE AoA notes — Silicon Labs BRD4185A (EFR32BG22)

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For readers who have not met baseband signals before. Everything below is trigonometry.

1 The received signal

The tag transmits the Constant Tone Extension (CTE) as a pure tone. At the anchor's antenna it arrives as a cosine with amplitude A , frequency f_r and some phase ϕ :

$$s_r(t) = A \cos(2\pi f_r t + \phi).$$

The receiver's local oscillator (LO) is tuned to the nominal channel frequency f_o . Because the tag's crystal and the anchor's crystal are never exactly equal, and because the CTE tone itself sits 250 kHz above the channel centre, the received frequency is not f_o but $f_r = f_o + \Delta f$. Substituting:

$$s_r(t) = A \cos(2\pi f_o t + 2\pi \Delta f t + \phi).$$

Everything we care about—the small offset Δf and the phase ϕ —is hidden inside the argument of a cosine oscillating at 2.4 GHz. Mixing is how we pull it out.

For compactness write the slow part of the argument as

$$\theta(t) = 2\pi \Delta f t + \phi, \quad \text{so that} \quad s_r(t) = A \cos(2\pi f_o t + \theta(t)).$$

2 Expand the cosine

Using $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$ with $\alpha = 2\pi f_o t$ and $\beta = \theta(t)$:

$$s_r(t) = A \left[\cos(2\pi f_o t) \cos \theta(t) - \sin(2\pi f_o t) \sin \theta(t) \right].$$

This already shows the structure: the signal is a fast cosine weighted by $\cos \theta$ plus a fast sine weighted by $\sin \theta$. The two weights are exactly the I and Q values we want. The mixer isolates them one at a time.

3 The I branch: multiply by $\cos(2\pi f_o t)$

$$s_r(t) \cos(2\pi f_o t) = A \left[\cos^2(2\pi f_o t) \cos \theta - \sin(2\pi f_o t) \cos(2\pi f_o t) \sin \theta \right].$$

Apply the double-angle identities $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$ and $\sin x \cos x = \frac{1}{2} \sin 2x$:

$$s_r(t) \cos(2\pi f_o t) = \frac{A}{2} \cos \theta + \frac{A}{2} \left[\cos(4\pi f_o t) \cos \theta - \sin(4\pi f_o t) \sin \theta \right].$$

The bracket is again $\cos(\alpha + \beta)$ with $\alpha = 4\pi f_o t$, so **before the low-pass filter** the mixer output is

$$s_r(t) \cos(2\pi f_o t) = \underbrace{\frac{A}{2} \cos \theta(t)}_{\text{baseband, slow}} + \underbrace{\frac{A}{2} \cos(4\pi f_o t + \theta(t))}_{\text{at } 2f_o \approx 4.8 \text{ GHz}} \quad (1)$$

The first term varies at Δf (hundreds of kHz); the second oscillates at twice the carrier. A low-pass filter with cut-off well below $2f_o$ removes the second term:

$$I(t) = \text{LPF}[s_r(t) \cos(2\pi f_o t)] = \frac{A}{2} \cos \theta(t) = I_o \cos(2\pi \Delta f t + \phi), \quad I_o = \frac{A}{2}.$$

4 The Q branch: multiply by $\sin(2\pi f_o t)$

$$s_r(t) \sin(2\pi f_o t) = A \left[\sin(2\pi f_o t) \cos(2\pi f_o t) \cos \theta - \sin^2(2\pi f_o t) \sin \theta \right].$$

With $\sin x \cos x = \frac{1}{2} \sin 2x$ and $\sin^2 x = \frac{1}{2} (1 - \cos 2x)$:

$$s_r(t) \sin(2\pi f_o t) = \frac{A}{2} \sin(4\pi f_o t) \cos \theta - \frac{A}{2} \sin \theta + \frac{A}{2} \cos(4\pi f_o t) \sin \theta.$$

Group the two fast terms; they form $\sin(\alpha + \beta)$ with $\alpha = 4\pi f_o t$:

$$s_r(t) \sin(2\pi f_o t) = \underbrace{-\frac{A}{2} \sin \theta(t)}_{\text{baseband, slow}} + \underbrace{\frac{A}{2} \sin(4\pi f_o t + \theta(t))}_{\text{at } 2f_o} \quad (2)$$

After the low-pass filter:

$$Q(t) = \text{LPF}[s_r(t) \sin(2\pi f_o t)] = -\frac{A}{2} \sin \theta(t) = Q_o \sin(2\pi \Delta f t + \phi), \quad Q_o = -\frac{A}{2}.$$

A note on the sign. Mixing with $+\sin(2\pi f_o t)$ gives $Q_o = -A/2$. Most receivers, including the EFR32, define the Q-branch LO as $-\sin(2\pi f_o t)$ (equivalently, the complex LO is $e^{-j2\pi f_o t} = \cos -j \sin$), which flips the sign to $Q_o = +A/2$. The choice is a hardware convention; it decides whether a positive Δf makes the I/Q phasor rotate counter-clockwise or clockwise. In our data the +250 kHz CTE tone shows up as a phase that *decreases* by about 74° per microsecond, i.e. as $\Delta f \approx -206 \text{ kHz}$. That negative sign is this convention at work, and it is the reason the analysis never assumes the sign of Δf but always measures it.

5 Put I and Q together: the complex baseband sample

Taking the receiver's convention $Q_o = +A/2$:

$$z(t) = I(t) + jQ(t) = \frac{A}{2} \left[\cos \theta(t) + j \sin \theta(t) \right] = \frac{A}{2} e^{j\theta(t)} = \frac{A}{2} e^{j(2\pi \Delta f t + \phi)}.$$

The 2.4 GHz carrier is gone. What remains is a phasor of constant length $A/2$ that rotates at the offset frequency Δf , starting from the phase ϕ .

6 Sampling, and what the CFO estimator measures

The ADC samples $z(t)$ every $T_s = 1 \mu\text{s}$ during the reference period:

$$z_k = z(kT_s) = \frac{A}{2} e^{j(2\pi \Delta f k T_s + \phi)}, \quad k = 0, 1, \dots, 7.$$

Two consecutive samples differ only by one step of rotation:

$$\overline{z_k} z_{k+1} = \frac{A^2}{4} e^{-j(2\pi \Delta f k T_s + \phi)} e^{j(2\pi \Delta f (k+1) T_s + \phi)} = \frac{A^2}{4} e^{j 2\pi \Delta f T_s}.$$

The unknown phase ϕ cancels. The angle of this product is the phase advance per microsecond, and dividing by $2\pi T_s$ gives Δf directly:

$$\Delta f = \frac{\angle(\overline{z_k} z_{k+1})}{2\pi T_s}.$$

Seven such products are available from the eight reference samples. They are normalised to unit length and averaged (a *circular mean*, so that angles near $\pm 180^\circ$ do not cancel):

$$\rho = \frac{1}{7} \sum_{k=1}^7 \frac{\overline{z_k} z_{k+1}}{|\overline{z_k} z_{k+1}|}, \quad \hat{\Delta f} = \frac{\angle \rho}{2\pi T_s}.$$

$|\rho|$ (the mean resultant length, γ) is a free quality indicator: 1 when the seven phase steps agree exactly, lower when the reference period is noisy. The $1 \mu\text{s}$ spacing gives an unambiguous range of $\pm 500 \text{ kHz}$, which comfortably contains $\pm 250 \text{ kHz}$ plus the crystal offset.

Worked example (Dataset19, set1, packet 0)

Tag 84:2e:14:31:ba:9e, channel 24, RSSI -58 dBm .

k	I_k	Q_k	$ z_k $	$\angle z_k$
1	-106	50	117.2	154.7°
2	19	116	117.5	80.7°
3	117	12	117.6	5.9°
4	44	-108	116.6	-67.8°
5	-94	-71	117.8	-142.9°
6	-95	69	117.4	144.0°
7	41	109	116.5	69.4°
8	114	-9	114.4	-4.5°

The seven conjugate products have angles $-74.05^\circ, -74.84^\circ, -73.69^\circ, -75.10^\circ, -73.06^\circ, -74.62^\circ, -73.90^\circ$. Their circular mean is

$$\rho = 0.2726 - 0.9621j, \quad |\rho| = 0.99993, \quad \angle \rho = -74.18^\circ,$$

so

$$\hat{\Delta f} = \frac{-74.18^\circ}{360^\circ \times 1 \mu\text{s}} = -206.06 \text{ kHz}.$$

De-rotating the eight samples by $\bar{\rho}^k$ brings all their phases to $154.5^\circ \pm 0.36^\circ$: the residual is the phase-noise floor of this packet.

7 Why this matters for angle of arrival

When the receiver later switches antennas every $2\text{ }\mu\text{s}$, the phase of z keeps rotating at Δf regardless of which antenna is connected. The phase difference between two antennas therefore contains a geometric part (what we want) plus $2\pi \Delta f \cdot 2\text{ }\mu\text{s} \approx -148^\circ$ per switch (what we must remove). Knowing Δf from the reference period lets us multiply each sample by $e^{-j2\pi\Delta f t}$ and undo the rotation, leaving only the geometry. Because Δf contains the tag's own crystal offset it is different for every tag and drifts with temperature, so it must be measured on every packet and never copied from one tag to another.